



# Cross-Document Validation using Document Intelligence

Milestone 2 : Data Collection and Data Preparation

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# Milestone Objective

**Prepare datasets required to evaluate a system that can perform :**

- Document entity extraction
- Cross-document validation
- Semantic equivalence detection
- Numeric normalization
- Conflict detection

**Two datasets were prepared to support these tasks:**

1. FUNSD Dataset
2. Healthcare SBC & Benefit Grid Dataset

An illustration of a white document with a blue magnifying glass over the date field. The document has a blue header bar, a light blue box containing 'Name: John Doe', and an orange box containing 'Date: 12/05/2021'. The magnifying glass is blue and is positioned over the date. The background is a blue gradient with a white circular shape on the left.

## Document Entity Extraction

# Dataset Overview

Two complementary datasets were used:

## 1. FUNSD Dataset

- Form understanding
- Key-value extraction
- Document entity recognition

## 2. SBC & Benefit Grid Dataset

- Cross-document validation
- Coverage comparison
- Cost-sharing verification

Together they simulate real-world document validation workflows.

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FILTER SCORES

Brand: RALEIGH (BELAIR portion not tested) Project #: 1969-105  
Commercial: LAKE - NEW PACK :40 (with BELAIR Badminton :20) Sample: 336  
Code #: BW-RT-69-98 PM6 Base: (234)  
Supplier: AUDIENCE STUDIES

TEST DATES

L. Angeles: 8/5 and 6  
Chicago: 8/8

PM6  
SCORES

COMMENTS

Overall 1.7

CITY  
Los Angeles 0.0  
Chicago 3.3

SEX  
Male 0.0  
Female 3.3

AGE  
16-25 0.0  
26-35 0.0  
36-45 0.0  
46 & Over 9.3

35 & Under 0.0  
36 & Over 5.0

This commercial was tested in color.

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# FUNSD Dataset

FUNSD (Form Understanding in Noisy Scanned Documents)

Key characteristics:

- Scanned form documents
- Layout-aware annotations
- Key-value entity pairs
- Real-world document noise

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# FUNSD DATA PREPARATION

## Steps performed:

1. Selected sample documents from FUNSD dataset
2. Identified logical document sections
3. Defined entity groups within each section

## Example entities:

- Name
- Address
- Date
- Numeric values
- Identifiers

Prepared dataset for entity extraction experiments.

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# FUNSD Data Augmentation

To simulate real-world document variations:

- Several modifications were introduced:
- Synonym replacement
- Numeric format variations
- Layout modifications
- OCR-style noise

Examples:

- 11% → 0.11
- Interest Rate → Applicable Interest

**Purpose:** Test robustness of entity extraction models.

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## FILTER SCORES

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Code #: BW-RT-69-98 PM6 Base: (234)  
Supplier: AUDIENCE STUDIES

## TEST DATES

L. Angeles: 8/5 and 6  
Chicago: 8/8

## PM6 SCORES

|             |     |
|-------------|-----|
| Overall     | 1.7 |
| CITY        |     |
| Los Angeles | 0.0 |
| Chicago     | 3.3 |
| SEX         |     |
| Male        | 0.0 |
| Female      | 3.3 |
| AGE         |     |
| 16-25       | 0.0 |
| 26-35       | 0.0 |
| 36-45       | 0.0 |
| 46 & Over   | 9.3 |
| 35 & Under  | 0.0 |
| 36 & Over   | 5.0 |

## COMMENTS

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# SBC Dataset

Summary of Benefits and Coverage (SBC) document will help you choose a health [plan](#). The SBC shows you how you and the [plan](#) would pay for covered health care services. NOTE: Information about the cost of this [plan](#) (called the [premium](#)) will be provided separately. For a summary. For more information about your coverage, or to get a copy of the complete terms of coverage, [insert contact information]. For common terms, such as [allowed amount](#), [balance billing](#), [coinsurance](#), [copayment](#), [deductible](#), [provider](#), or other underlined terms see the Glossary at [www.\[insert\].com](#) or call 1-800-[insert] to request a copy.

Summary of Benefits and Coverage (SBC) documents

Healthcare insurance documents containing:

- Coverage information
- Cost-sharing details
- Healthcare service descriptions

Typical document size: 2 to 11 pages

These documents serve as Doc A in the dataset.

| Q                                                                                                    | Answers                                                                                                                                                                       | Why This Matters:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is the deductible?                                                                              | \$500/Individual or \$1,000/family                                                                                                                                            | Generally, you must pay all of the costs from providers up to the <a href="#">deductible</a> amount before this <a href="#">plan</a> begins to pay. If you have other family members on the <a href="#">plan</a> , each family member must meet their own individual <a href="#">deductible</a> until the total amount of <a href="#">deductible</a> expenses paid by all family members meets the overall family <a href="#">deductible</a> .                                                                                                                                                                                                            |
| What services are covered before you meet your deductible?                                           | Yes. <a href="#">Preventive care</a> and primary care services are covered before you meet your <a href="#">deductible</a> .                                                  | This <a href="#">plan</a> covers some items and services even if you haven't yet met the <a href="#">deductible</a> amount. But a <a href="#">copayment</a> or <a href="#">coinsurance</a> may apply. For example, this <a href="#">plan</a> covers certain <a href="#">preventive services</a> without <a href="#">cost sharing</a> and before you meet your <a href="#">deductible</a> . See a list of covered <a href="#">preventive services</a> at <a href="https://www.healthcare.gov/coverage/preventive-care-benefits/">https://www.healthcare.gov/coverage/preventive-care-benefits/</a> .                                                       |
| What are the specific deductibles for prescription drug coverage and occupational therapy services?  | Yes. \$300 for <a href="#">prescription drug coverage</a> and \$300 for occupational therapy services.                                                                        | You must pay all of the costs for these services up to the specific <a href="#">deductible</a> amount before this <a href="#">plan</a> begins to pay for these services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| What is the out-of-pocket limit?                                                                     | For <a href="#">network providers</a> \$2,500 individual / \$5,000 family; for <a href="#">out-of-network providers</a> \$4,000 individual / \$8,000 family                   | The <a href="#">out-of-pocket limit</a> is the most you could pay in a year for covered services. If you have other family members in this <a href="#">plan</a> , they have to meet their own <a href="#">out-of-pocket limits</a> until the overall family <a href="#">out-of-pocket limit</a> has been met.                                                                                                                                                                                                                                                                                                                                             |
| What are the copayments, premiums, balance-billing charges, and health care this plan doesn't cover? | <a href="#">Copayments</a> for certain services, <a href="#">premiums</a> , <a href="#">balance-billing</a> charges, and health care this <a href="#">plan</a> doesn't cover. | Even though you pay these expenses, they don't count toward the <a href="#">out-of-pocket limit</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| What is the provider network?                                                                        | Yes. See <a href="#">www.[insert].com</a> or call 1-800-[insert] for a list of <a href="#">network providers</a> .                                                            | This <a href="#">plan</a> uses a provider <a href="#">network</a> . You will pay less if you use a <a href="#">provider</a> in the plan's <a href="#">network</a> . You will pay the most if you use an <a href="#">out-of-network provider</a> , and you might receive a bill from a <a href="#">provider</a> for the difference between the provider's charge and what your <a href="#">plan</a> pays ( <a href="#">balance billing</a> ). Be aware, your <a href="#">network provider</a> might use an <a href="#">out-of-network provider</a> for some services (such as lab work). Check with your <a href="#">provider</a> before you get services. |
| Is a referral to a specialist required?                                                              | Yes.                                                                                                                                                                          | This <a href="#">plan</a> will pay some or all of the costs to see a <a href="#">specialist</a> for covered services but only if you have a <a href="#">referral</a> before you see the <a href="#">specialist</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                      |

# SBC Data Collection

SBC documents were collected by searching for:

- Highmark PPO insurance plans

Challenges encountered:

- Duplicate documents
- Different formatting styles

Solutions:

Searching across plan tiers

- Gold
- Silver
- Bronze

Using multiple plan identifiers

Total collected: 20 SBC documents



# Benefit Grid Creation

To standardize SBC information: A Benefit Grid template was created.

The grid includes columns for:

- Service Name
- In-Network Cost
- Out-of-Network Cost
- Notes

Example services:

- Doctor visits
- Emergency care
- Hospital stay
- Prescription drugs

Each SBC was converted into a Benefit Grid document (Doc B).

# Document Pair Structure

Each dataset sample consists of:

Doc A: SBC Document

Doc B: Benefit Grid

Example relationship:

SBC Document → Detailed explanation

Benefit Grid → Structured summary

This structure enables cross-document validation.

# Data Augmentation (Benefit Grid)

Controlled modifications were applied to simulate real-world inconsistencies.

Out of 20 benefit grids:

- 12 kept unchanged
- 8 intentionally modified

Modifications were applied using PDF editing tools.

# Augmentation Types

## Semantic Changes

Example: "No charge" → "Covered in full"

## Numeric Format Changes

Example: "\$6,550" → "6550"

## Value Conflicts

Example: "\$525 copay" → "\$400 copay"

## Coverage Changes

Example: "\$30 copay" → "Fully covered"

# Ground Truth JSON Creation

Ground truth JSON files were created for augmented document pairs.

Each JSON contains:

- Entity name
- Value in Doc A
- Value in Doc B
- Normalized value
- Validation type
- Match or mismatch result

Also includes perturbation logs to track modifications.

# Validation Scenarios Covered

**Dataset supports multiple validation cases:**

1. Exact Match
2. Semantic Equivalence
3. Numeric Normalization
4. OCR Noise Handling
5. Conflict Detection
6. Coverage Change Detection

These scenarios represent real-world document inconsistencies.

# Dataset Output

## Final dataset produced:

| Dataset Component              | Count    |
|--------------------------------|----------|
| FUNSD Samples                  | Multiple |
| SBC Documents (Doc A)          | 20       |
| Benefit Grid Documents (Doc B) | 20       |
| Ground Truth JSON Files        | 8        |

This dataset supports both:

- Entity extraction
- Cross-document validation

# Challenges Faced

## **During dataset preparation:**

- Duplicate Documents Many identical SBC files online.
- Complex Tables Multiple network tiers in SBC tables.
- Formatting Issues PDF extraction caused misaligned columns.
- Ambiguous Service Descriptions Some rows combined multiple services.
- Manual verification was performed to resolve these issues.

# Team Contributions

## **Karthik**

Researched the validation use case, Identified FUNSD and SBC datasets, Designed augmentation scenarios

## **Mayank**

Worked on FUNSD data augmentation, Created modified document samples

## **Ayush**

Performed SBC and Benefit Grid augmentation

Generated ground truth JSON files

## **Mallesha**

Documentation.

# Milestone Summary

Milestone 2 successfully achieved:

- ✓ Dataset research and selection
- ✓ Data collection from multiple sources
- ✓ Document standardization
- ✓ Controlled data augmentation
- ✓ Ground truth generation

Prepared dataset enables:

- ✓ Entity extraction evaluation
- ✓ Cross-document validation experiments

**Thank You :)**